

A Sahni-style Difficulty Measure for Ubongo

From the 0-1 Knapsack Problem to geometric exact cover

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The board game Ubongo asks a player to fill a printed outline exactly, using a prescribed set of polyomino tiles, with no gaps and no overlaps. Computer scientists classify this as a variant of the Exact Cover Problem and a two-dimensional Bin-Packing / Knapsack problem: a fixed "capacity" (the outline) must be packed with indivisible items (the pieces). This note transfers the Sahni-k difficulty measure, originally defined for the 0-1 knapsack problem, to Ubongo, and records two complementary measures used by the live game to grade its puzzles.

1. The knapsack / Ubongo correspondence

Feature	0-1 Knapsack Problem	Ubongo
Objective	Maximise the value of items packed into a bag without exceeding a weight limit.	Fill a shaded board region exactly with geometric pieces - no overlap, no spill.
Capacity	A scalar weight budget (e.g. 15 kg).	The exact cell-outline printed on the card (an area budget and a shape).
Items / pieces	Each item has a weight and a value.	A set of polyomino (tromino / tetromino / pentomino) tiles, each of fixed area.
Indivisibility	An item is taken whole (1) or not at all (0); no fractions.	Pieces cannot be cut; each is placed whole, with rotation / reflection.

The essential difference is dimensionality: knapsack feasibility is governed by a single scalar (total weight $\leq$ capacity), whereas Ubongo feasibility is governed by 2-D geometry (placements must tile a specific shape). Ubongo is therefore most precisely an instance of geometric exact cover, solvable with Knuth's Algorithm X / Dancing Links.

2. The Sahni-k difficulty of a knapsack instance

Sahni's (1975) approximation scheme S_k for the 0-1 knapsack works as follows: enumerate every subset of at most k items, force that subset into the bag, then complete the packing greedily in non-increasing value-to-weight ratio order; return the best packing found over all such subsets. As k grows the guarantee improves (relative error at most $1/(k+1)$), and at $k = n$ the method is exact.

The Sahni-difficulty of an instance is the smallest k for which S_k returns the optimum.

Intuitively, it is the fewest decisions one must commit by hand before a myopic greedy rule finishes the job correctly. A value $k = 0$ means greedy is already optimal (easy); a large k means greedy is repeatedly led into traps and substantial search is unavoidable (hard). Two ingredients make the measure work: (i) a canonical greedy rule, and (ii) the notion of "minimum forced decisions" to reach the target. Both port to Ubongo.

3. A Sahni-style difficulty for Ubongo

3.1 Instance

An instance is a region R of N cells together with a multiset of polyominoes $P = \{p_1, \dots, p_m\}$ whose areas sum to N . A solution (tiling) places every piece - choosing an orientation (rotations and reflections allowed) and a translation - so that the placements partition R exactly. Because the areas sum to N , every solution uses all m pieces exactly once: this is exact cover.

3.2 Canonical greedy H (the analogue of ratio order)

Place pieces largest-area-first (ties broken by a fixed piece index); for each piece, take the first feasible placement under a fixed top-to-bottom, left-to-right cell scan and a fixed orientation order. The rule is deterministic and never backtracks. It either tiles R (success) or reaches a cell that no remaining piece can legally cover (a dead end) - mirroring knapsack greedy producing a feasible but sub-optimal pack.

3.3 Definition

Let $H\text{-completion}(S)$ denote the result of running H after pre-placing a set S of piece-placements. Write $k(R,P)$ for the difficulty. Then:

$$k(R,P) = \min \text{ over tilings } T \text{ of } \min \{ |S| : S \text{ subset of } T, \text{ and } H\text{-completion}(S) \text{ yields a full tiling} \}.$$

In words: k is the fewest correctly-placed pieces that must be revealed before a naive greedy player can finish the board without ever getting stuck. In the live game this is literally the minimum number of times the Hint button must be pressed before greedy play coasts home.

3.4 Properties (parallel to Sahni-k)

- $k = 0$: the greedy rule tiles the board outright - trivial.
- $k = m$: every piece is a trap; the seed must contain the whole solution - pure search.
- k forms a hierarchy $0, 1, \dots, m$ and is always finite, because pinning all m pieces of any known tiling completes trivially (so $k \leq m$).
- Computing k is itself the Sahni-style subset enumeration; for puzzle sizes with $m \leq 5$ it is negligible (at most $2^m \leq 32$ greedy runs per tiling).

3.5 Algorithm

```
function sahnIDifficulty(R, P):
  if H-completion(EMPTY) tiles R:      return 0      # k = 0
  best <- m
  for each tiling T of (R, P):          # via Algorithm X / DLX
    for k = 1 .. best-1:
      for each subset S of T with |S| = k:
        if H-completion(S) tiles R: best <- k; break
  if best == 1: return 1                  # cannot beat k=1 once k != 0
  return best
```

Short-circuiting (test $k = 0$ first; stop at the first tiling that needs a single hint) keeps the computation fast in practice; solution enumeration is capped for robustness.

4. Two complementary measures

k captures trap structure (the Sahni spirit). Two further signals capture scarcity and human search effort, and are solver-light:

Measure	What it captures	Cost
k (Sahni analogue)	Minimum forced hints before greedy succeeds.	$O(2^m \cdot N)$
Solution count $ T $	Scarcity: a unique tiling is hardest; many tilings are forgiving.	One DLX count.
Search nodes / backtracks	Proxy for the effort a backtracking solver (or human) expends.	One solve.

The three agree at the extremes and diverge in the middle - the Ubongo version of "where are the

hard knapsack problems": difficulty peaks when the number of tilings is small but positive and the pieces interlock (high k), not merely when the board is large.

5. Use in the live game

Every board uses the full piece set - one tromino, three tetrominoes and four pentominoes, 35 cells in total - so the same eight pieces are always shown; difficulty varies only the shape of the outline. The generator builds each board from a known tiling, so a solution always exists and k , $|T|$ are computed exactly and cheaply. The game rejection-samples outlines until the measured k matches the tier (keeping the closest candidate as a fallback), and shows k and the tiling count beneath the board.

With all eight pieces the measured difficulty spreads naturally over $k = 1..4$, giving a reachable, monotone ladder (here $m = 8$ is fixed, so difficulty is carried entirely by board geometry):

Tier	Pieces	Target k	Typical tilings $ T $
Easy	8 (all)	$k \leq 1$	a few
Medium	8 (all)	$k = 2$	1 - 2
Hard	8 (all)	$k = 3$	about 1
Expert	8 (all)	$k \geq 4$	1 (often unique)

References

- S. Sahni, "Approximate algorithms for the 0/1 knapsack problem," Journal of the ACM 22(1):115-124, 1975.
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